Project: ClinicFlow - Outpatient Clinic and Medical Inventory Management System

1 Introduction

ClinicFlow is an integrated platform for outpatient clinics, small hospitals, and specialty practices to manage patient registration, appointments, clinical workflows, electronic medical records (EMR-lite), prescriptions, pharmacy and inventory, billing, reporting, and regulatory compliance. The system emphasizes deterministic clinical and administrative rules, audit-ready records, role-scoped access, and offline-capable workflows for point-of-care operations. Telemetry from medical devices is out of scope for the MVP.

Primary actors

- **Patient**: registers, books appointments, views invoices and prescriptions.
- **Reception / Front Desk**: manages registrations, appointments, and billing.
- **Clinician (Doctor/Nurse/Practitioner)**: documents encounters, prescribes medications, orders tests.
- **Pharmacist / Pharmacy Staff**: manages dispensing, inventory, and medication reconciliation.
- **Lab Technician**: records test results and sample tracking.
- **Finance Officer**: handles invoicing, payments, insurance claims, and reconciliations.
- **Clinic Manager**: oversees schedules, resource utilization, and compliance.
- **Administrator**: configures roles, services, tariffs, and integrations.
- **Compliance Officer / Auditor**: reviews audit trails and generates regulatory packages.

2 Module Overview

- Identity & Access Management
- Patient Registry & Appointments
- Clinical Encounters & EMR Lite
- Prescription Management & E-Prescribing (optional integrations)
- Pharmacy & Inventory Management
- Laboratory Orders & Results Management
- Billing, Insurance Claims & Payments
- Reporting, KPIs & Audit Packages
- Notifications, Reminders & Task Lists
- Integrations (Insurance, Labs, Payment Gateways, HIS/ERP)

3 Architecture Overview

- **Frontend**: Role-based web portals for clinicians, reception, pharmacy, finance; patient portal and mobile-friendly views for appointments and invoices.
- **Backend**: Modular services (Identity, Patient, Appointments, Encounters, Pharmacy, Lab, Billing, Reporting) with REST/GraphQL APIs and event streams for asynchronous tasks.
- **Database**: Relational DB (PostgreSQL) for authoritative records; JSONB for flexible clinical notes and order payloads.
- **Storage**: Encrypted object storage for scanned documents, attachments, and imaging reports; DB stores URIs and checksums.
- **Processing model**: Deterministic rules for appointment allocation, prescription checks (drug interactions, stock), billing rules, and insurance adjudication.
- **Deployment**: Cloud or on-premises; containerized services; centralized logging, encrypted backups, and role-scoped encryption keys.
- **Integrations**: Insurance clearinghouses, external labs (HL7/REST), payment gateways, national health registries, and pharmacy networks.

4 Module Wise Design

4.1 Identity & Access Management

Features

- Secure authentication, RBAC, MFA for privileged roles, session management, and append-only audit logs.
- Role scoping per clinic/department and emergency access (break-glass) with audit trail.

Entities

- **User** [UserID, Name, Role [Patient/Reception/Clinician/Pharmacy/Finance/Manager/Admin/Auditor], Email, Phone, Status, CreatedAt, UpdatedAt]
- **AuditLog** [AuditID, UserID FK, Action, ResourceType, ResourceID, DetailsJSON, Timestamp]

4.2 Patient Registry & Appointments

Features

- Patient registration, demographic management, identifiers (MRN), consent capture, and insurance details.
- Appointment booking with deterministic slot allocation, waitlist, cancellations, and reminders.

Entities

- **Patient** [PatientID, MRN, Name, DOB, Gender, ContactInfoJSON, AddressJSON, PrimaryContact, InsuranceID FK|null, Status, CreatedAt]
- **Appointment** [ApptID, PatientID FK, ClinicianID FK, Department, ServiceType, StartAt, EndAt, Status [Scheduled/CheckedIn/Completed/Cancelled/NoShow], CreatedByFK, CreatedAt]

4.3 Clinical Encounters & EMR Lite

Features

- Structured encounter templates (SOAP), problem lists, medication lists, allergies, vitals (manual entry), and deterministic clinical decision checks (allergy, duplicate therapy).
- Encounter versioning, signed notes, and clinician sign-off workflows.

Entities

- **Encounter** [EncounterID, PatientID FK, ClinicianID FK, VisitType, ChiefComplaint, VitalsJSON, NotesJSON, DiagnosesJSON, OrdersJSON, PrescriptionsJSON, StartAt, EndAt, Status, SignedByFK, SignedAt]
- **ProblemList** [ProblemID, PatientID FK, Code, Description, OnsetDate, Status, CreatedAt]

4.4 Prescription Management & E-Prescribing

Features

- Create prescriptions with deterministic checks (allergy, interactions, dose limits), e-prescribe to pharmacy networks (optional), and print/issue paper prescriptions.
- Prescription refill workflows and audit trail for controlled substances.

Entities

- **Prescription** [RxID, EncounterID FK, PatientID FK, ClinicianID FK, MedicationCode, Dosage, Frequency, DurationDays, Quantity, Repeats, Route, Notes, Status [Draft/Issued/Dispensed/Cancelled], IssuedAt]
- **MedicationMaster** [MedID, Code, Name, Formulation, Strength, ATCCode, ControlledFlag, Status]

4.5 Pharmacy & Inventory Management

Features

- Stock management, batch/lot tracking, expiry management, dispensing workflows, and stock reservation for prescriptions.
- Deterministic dispensing checks and pharmacy returns/refunds.

Entities

- **InventoryItem** [InventoryID, MedID FK, BatchNumber, Quantity, Unit, ExpiryDate, Location, CostPrice, Status]
- **DispenseRecord** [DispenseID, RxID FK, InventoryID FK, PatientID FK, DispensedByFK, Quantity, DispensedAt, Notes, Status]

4.6 Laboratory Orders & Results Management

Features

- Order lab tests, sample tracking, result ingestion (manual or HL7/REST), result flagging, and clinician notifications.
- Deterministic validation for critical values and reflex testing rules.

Entities

- **LabOrder** [LabOrderID, EncounterID FK, PatientID FK, OrderedByFK, TestsJSON, SampleID, CollectedAt, Status, ResultURI]
- **LabResult** [ResultID, LabOrderID FK, TestCode, Value, Units, ReferenceRangeJSON, Flag [Normal/High/Low/Critical], ReportedAt, ReportedBy]

4.7 Billing, Insurance Claims & Payments

Features

- Service catalog and tariff engine, invoice generation, insurance claim creation and adjudication, co-pay handling, refunds, and payment reconciliation.
- Deterministic claim rules and denial handling workflows.

Entities

- **ServiceItem** [ServiceID, Code, Description, Price, Department, BillingCategory, Status]
- **Invoice** [InvoiceID, PatientID FK, EncounterID FK|null, LineItemsJSON, Subtotal, Taxes, Discounts, TotalAmount, IssuedAt, DueDate, Status]
- **Payment** [PaymentID, InvoiceID FK, PatientID FK, Amount, Method, PaidAt, Status]
- **InsuranceClaim** [ClaimID, InvoiceID FK, PayerID FK, ClaimPayloadJSON, SubmittedAt, Status, ResponseJSON]

4.8 Reporting, KPIs & Audit Packages

Features

- Clinical and operational dashboards (utilization, wait times, revenue), regulatory reporting, and exportable audit packages for inspections.

Entities

- **Report** [ReportID, Scope [Clinical/Finance/Compliance], ParametersJSON, MetricsJSON, GeneratedByFK, GeneratedAt, ReportURI]
- **KPI** [KPIID, Name, Definition, Target, CurrentValue, ReportingPeriod]
- **AuditPackage** [PackageID, PeriodStart, PeriodEnd, ContentsJSON, GeneratedAt, PackageURI]

4.9 Notifications, Reminders & Task Lists

Features

- Appointment reminders, medication refill reminders, lab result notifications, task assignments for staff, and escalation rules.

Entities

- **Notification** [NotificationID, UserID FK|null, PatientID FK|null, EntityID, Message, Category, Channel [SMS/Email/Push], Severity, CreatedAt, ReadAt, Status]
- **Task** [TaskID, AssignedToFK, RelatedEntityID, Description, DueDate, Priority, CreatedAt, CompletedAt, Status]

4.10 Integrations and Adapters

Features

- Connectors for insurance clearinghouses, external labs (HL7/REST), payment gateways, national health registries, and pharmacy networks.
- Sandbox mode for integration testing and configurable mapping templates.

Entities

- **Adapter** [AdapterID, Type [Lab/Insurance/Payment/Pharmacy], ConfigJSON, CredentialsEncrypted, LastSyncAt, Status]

5 Deployment Strategy

- **Development**: Local developer environments with synthetic patient and appointment datasets; unit and integration tests for prescription checks and billing rules.
- **Staging**: Staging with masked production snapshots for UAT; integration tests with lab and insurance sandboxes.
- **Production**: Cloud or on-premises per regulatory needs; containerized services with Kubernetes; managed PostgreSQL with read replicas; encrypted object storage for documents.
- **Backups and DR**: Daily encrypted backups, point-in-time recovery, offsite snapshots, documented RTO/RPO, and periodic restore drills.

- **Security**: TLS for all endpoints, encryption at rest for PHI, strict RBAC, MFA for privileged roles, HSM for key management where required, and regular penetration testing.
- **Operational**: 24x7 support options, runbooks for critical clinical and billing incidents, and audit readiness procedures.
- **Privacy & Compliance**: Configurable retention and redaction policies to meet HIPAA, GDPR, and local health data regulations.

6 Database Design

- **User** [UserID PK, Name, Role, Email, Phone, Status, CreatedAt, UpdatedAt]
- **AuditLog** [AuditID PK, UserID FK, Action, ResourceType, ResourceID, DetailsJSON, Timestamp]
- **Patient** [PatientID PK, MRN UNIQUE, Name, DOB, Gender, ContactInfoJSON, AddressJSON, PrimaryContact, InsuranceID FK|null, Status, CreatedAt]
- **Appointment** [ApptID PK, PatientID FK, ClinicianID FK, Department, ServiceType, StartAt, EndAt, Status, CreatedByFK, CreatedAt]
- **Encounter** [EncounterID PK, PatientID FK, ClinicianID FK, VisitType, ChiefComplaint, VitalsJSON, NotesJSON, DiagnosesJSON, OrdersJSON, PrescriptionsJSON, StartAt, EndAt, Status, SignedByFK, SignedAt]
- **ProblemList** [ProblemID PK, PatientID FK, Code, Description, OnsetDate, Status, CreatedAt]
- **MedicationMaster** [MedID PK, Code, Name, Formulation, Strength, ATCCode, ControlledFlag, Status]
- **Prescription** [RxID PK, EncounterID FK, PatientID FK, ClinicianID FK, MedID FK, Dosage, Frequency, DurationDays, Quantity, Repeats, Route, Notes, Status, IssuedAt]
- **InventoryItem** [InventoryID PK, MedID FK, BatchNumber, Quantity, Unit, ExpiryDate, Location, CostPrice, Status]
- **DispenseRecord** [DispenseID PK, RxID FK, InventoryID FK, PatientID FK, DispensedByFK, Quantity, DispensedAt, Notes, Status]
- **LabOrder** [LabOrderID PK, EncounterID FK, PatientID FK, OrderedByFK, TestsJSON, SampleID, CollectedAt, Status, ResultURI]
- **LabResult** [ResultID PK, LabOrderID FK, TestCode, Value, Units, ReferenceRangeJSON, Flag, ReportedAt, ReportedBy]
- **ServiceItem** [ServiceID PK, Code, Description, Price, Department, BillingCategory, Status]
- **Invoice** [InvoiceID PK, PatientID FK, EncounterID FK|null, LineItemsJSON, Subtotal, Taxes, Discounts, TotalAmount, IssuedAt, DueDate, Status]

- **Payment** [PaymentID PK, InvoiceID FK, PatientID FK, Amount, Method, PaidAt, Status]
- **InsuranceClaim** [ClaimID PK, InvoiceID FK, PayerID FK, ClaimPayloadJSON, SubmittedAt, Status, ResponseJSON]
- **Report** [ReportID PK, Scope, ParametersJSON, MetricsJSON, GeneratedByFK, GeneratedAt, ReportURI]
- **KPI** [KPIID PK, Name, Definition, Target, CurrentValue, ReportingPeriod]
- **AuditPackage** [PackageID PK, PeriodStart, PeriodEnd, ContentsJSON, GeneratedAt, PackageURI]
- **Notification** [NotificationID PK, UserID FK|null, PatientID FK|null, EntityID, Message, Category, Channel, Severity, CreatedAt, ReadAt, Status]
- **Task** [TaskID PK, AssignedToFK, RelatedEntityID, Description, DueDate, Priority, CreatedAt, CompletedAt, Status]
- **Adapter** [AdapterID PK, Type, ConfigJSON, CredentialsEncrypted, LastSyncAt, Status]

Design notes

- Enforce referential integrity and soft-delete patterns for PHI retention; use cascading rules carefully to preserve audit trails.
- Unique constraints: MRN per clinic; Appointment unique on (ClinicianID, StartAt, EndAt) to prevent double booking.
- Indexes: (PatientID), (ClinicianID, StartAt), (InvoiceID, Status), (LabOrderID, Status).
- Store large clinical attachments in encrypted object storage; record URIs and SHA-256 hashes for tamper evidence.
- Use JSONB for flexible clinical notes and order payloads; add GIN indexes for common keys.
- Maintain append-only AuditLog and immutable snapshots for Encounter and Prescription records for medico-legal compliance.
- Partition high-growth tables (Appointment, Encounter, LabResult, Invoice) by date or clinic for scale.
- Implement optimistic concurrency controls for prescription issuance and inventory adjustments to prevent race conditions.

7 User Interface Design

- **Reception Portal**: patient search/registration, appointment booking, check-in/out, and quick billing.
- **Clinician Workspace**: encounter templates, problem list, medication reconciliation, order entry, and signed notes.

- **Pharmacy Console**: prescription queue, stock reservation, dispensing, batch/expiry alerts, and returns.
- **Lab Dashboard**: sample tracking, result entry, critical value alerts, and result distribution.
- **Finance Dashboard**: invoice generation, claim submission, payment reconciliation, and revenue reports.
- **Patient Portal**: appointment booking, visit summaries, prescriptions, invoices, and secure messaging.
- **Manager Dashboard**: utilization, wait times, clinician productivity, inventory health, and compliance tasks.
- **Reporting Portal**: scheduled/ad-hoc reports, KPI dashboards, and audit package exports.
- **Admin Panel**: user/role management, service catalog, tariff editor, adapter configuration, and system health.

8 Non-Functional Requirements

- **Performance**: patient lookup and appointment booking under 1 second; encounter save and prescription checks within 500ms.
- **Security & Privacy**: TLS for all traffic, encryption at rest for PHI, strict RBAC, MFA for privileged roles, HSM for keys where required, and regular security audits.
- **Scalability**: horizontal scaling for API and web layers; partitioning for multi-clinic deployments.
- **Availability**: high availability for appointment and prescription services; multi-AZ deployment recommended.
- **Reliability**: durable, idempotent operations for billing and prescription issuance; offline capture for reception/clinician notes with reconciliation.
- **Consistency**: transactional safeguards for inventory and billing operations to prevent inconsistencies.
- **Maintainability**: modular microservices, API versioning, CI/CD pipelines, automated DB migrations, and runbooks for clinical incidents.
- **Observability**: centralized logging, tracing, and metrics for appointment throughput, prescription latency, claim success rates, and inventory alerts.
- **Compliance & Auditability**: append-only audit logs, versioned clinical records, and exportable audit packages for regulatory inspections.
- **Usability**: clinician UI optimized for rapid documentation; patient portal optimized for simple booking and payments.
- **Interoperability**: adapters for HL7/REST lab integrations, insurance clearinghouses, pharmacy networks, and payment gateways.

9 Assumptions & Constraints

- **No Device Telemetry**: direct medical device telemetry (e.g., bedside monitors) is out of scope for MVP; vitals are entered manually or via machinery uploads where available.
- **Deterministic Rules**: clinical checks, appointment allocation, and billing rules are deterministic; no AI decisioning in MVP.
- **Data Residency & Compliance**: deployment model (cloud vs on-prem) and data residency must be specified to meet local health regulations.
- **Integration Variability**: external lab and insurance APIs vary by region; adapter development and certification may be required.
- **PHI Handling**: customers must provide secure network segmentation and operational policies; ClinicFlow enforces technical controls but cannot manage local operational security.
- **Volume Assumptions**: designed for small to medium clinics; enterprise hospital scale requires additional partitioning and capacity planning.
- **Training & Adoption**: clinicians, reception, pharmacy and finance staff require onboarding and SOP alignment.
- **Retention Policies**: legal retention and redaction policies must be configured per jurisdiction.
- **Offline Support**